TrES-5b

R-0724

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_250528` · created 2026-07-28T00:13:18+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460823.8007 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.018
Transit depth (Rp/Rs)^2	2.54 +/- 0.58 [%]
Orbital Inclination (inc)	84.49 +/- 1.99
Airmass coefficient 1 (a1)	1.064 +/- 0.027
Airmass coefficient 2 (a2)	-0.037 +/- 0.011
Scatter in the residuals of the lightcurve fit is	2.54 %
Best Comparison Star	#2 - [289, 417]
Optimal Method	PSF photometry
Transit Duration (day)	0.073 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.018 vs 0.142 (deviation 0.84σ)
Duration fit vs published	0.073 d vs 0.0758 d (deviation 0.2σ)
Mid-time O–C	-11.0 min
Depth SNR	7.5
Residual scatter	2.54 %

Light curve

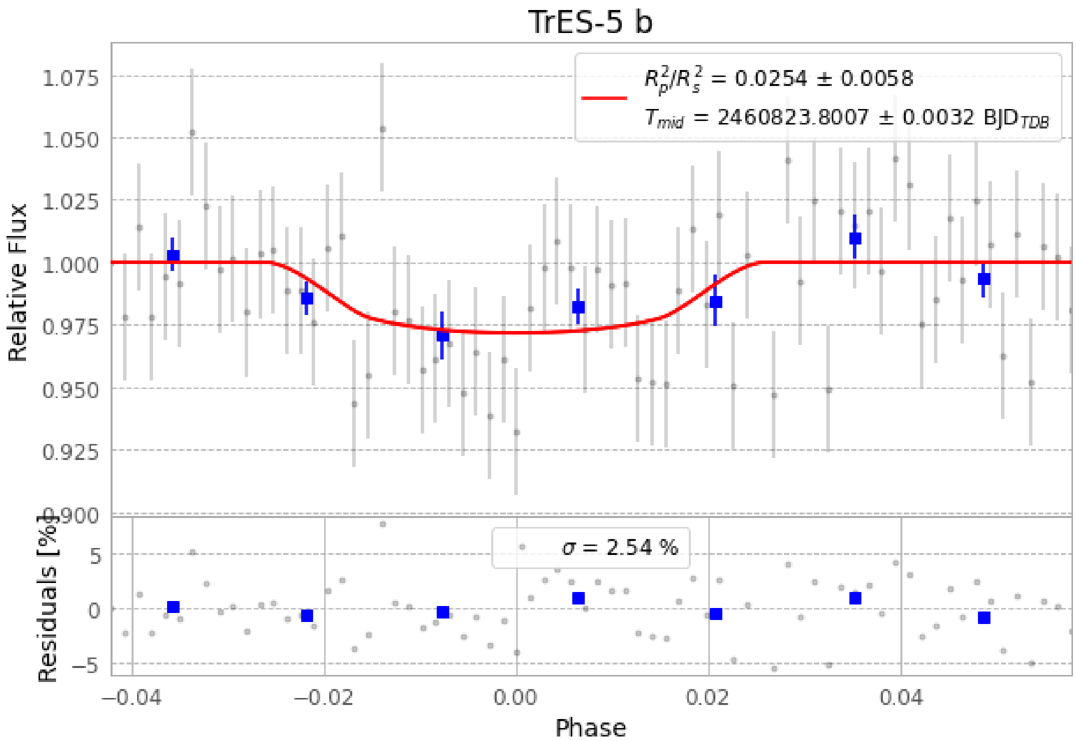

Figure 1 — FinalLightCurve_TrES-5 b_2025-05-27.png (SHA-256 5bef7c3a0b32b3a5…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 173], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 efe45df277972810…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

72 FITS frames (47 MB), `TRES-5250528054215.FITS` → `TRES-5250528091515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250528.log (0aea6b7d5b53…), FinalLightCurve_TrES-5 b_2025-05-27.png (5bef7c3a0b32…), FinalLightCurve_TrES-5 b_2025-05-27.csv (c5c6f63cad5b…)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 00:13Z.